

Semiconductors: New Technologies and the Value Chain

Module 2: Semiconductor materials and their properties

Live Session

by Dr Matthew Tang

Learning Outcomes

By the end of this module you will be able to:

- Classify the range of materials to manufacture integrated circuit: semiconductor – silicon, germanium; insulator and conductor materials like silicon dioxide, copper.
- Explain the preparation of semiconductor materials and the photolithographic process in semiconductor manufacturing.
- Explore the engineering of new semiconductor materials.

A warm-up question

Will Silicon be replaced soon?

- What makes Silicon a preferable choice for semiconductor chips?
- Is it related to Moore's law?
- What difficulties have to overcome in manufacturing?

Flexible Chips/Electronics

- For wearable technology, clothing authentication, and even in parcel labels to track and trace items
- Substrate: flexible thin film (e.g. PET)
- Semiconductor: amorphous Indium Gallium Zinc Oxide (a-IGZO) – flexible, fully transparent transistor

Why is the world investing so much in semiconductors?

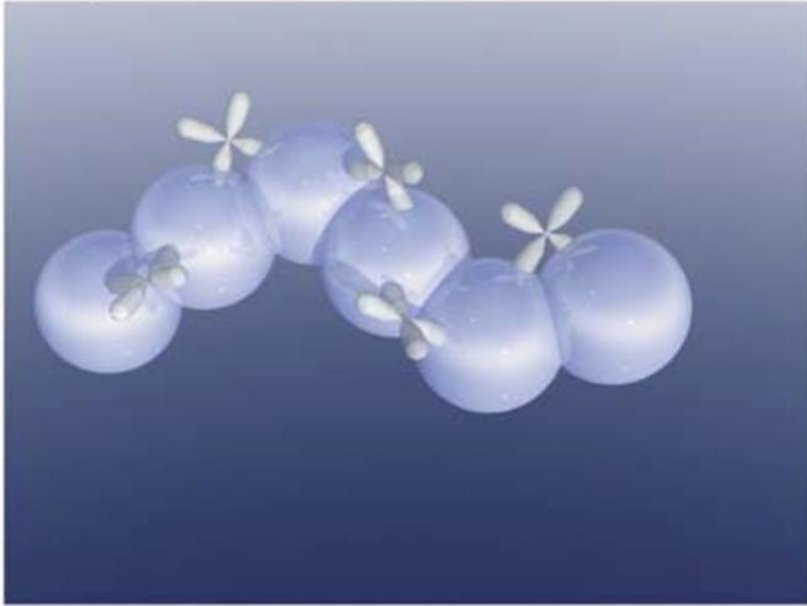
🕒 12 January 2024

[1] <https://www.bbc.co.uk/news/business-67576918>



IGZO: Properties

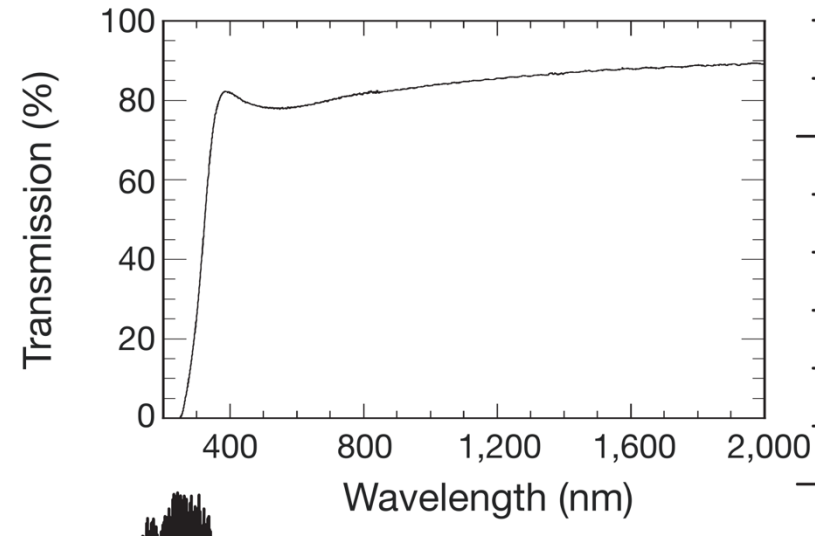
(1) atomic structure



**Room-temperature fabrication
of transparent flexible thin-film
transistors using amorphous
oxide semiconductors**

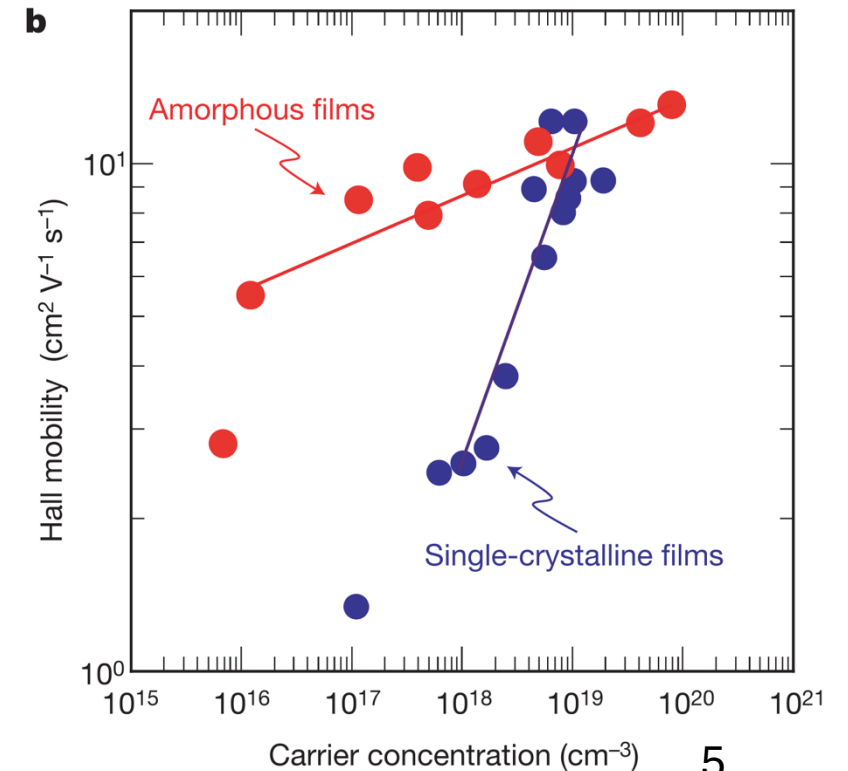
Kenji Nomura¹, Hiromichi Ohta¹, Akihiro Takagi², Toshio Kamiya^{1,2},
Masahiro Hirano¹ & Hideo Hosono^{1,2,3}

(2) optical
transmission



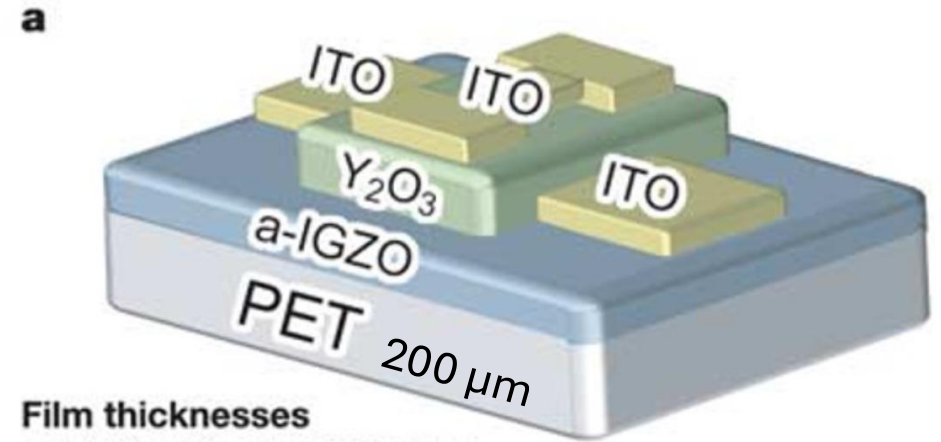
(3) electrical properties

Bandgap $E_g \sim 3.0$ eV
Conductivity $< 10^{-5}$ S cm⁻¹
Mobility: ~ 10 cm² V⁻¹ s⁻¹



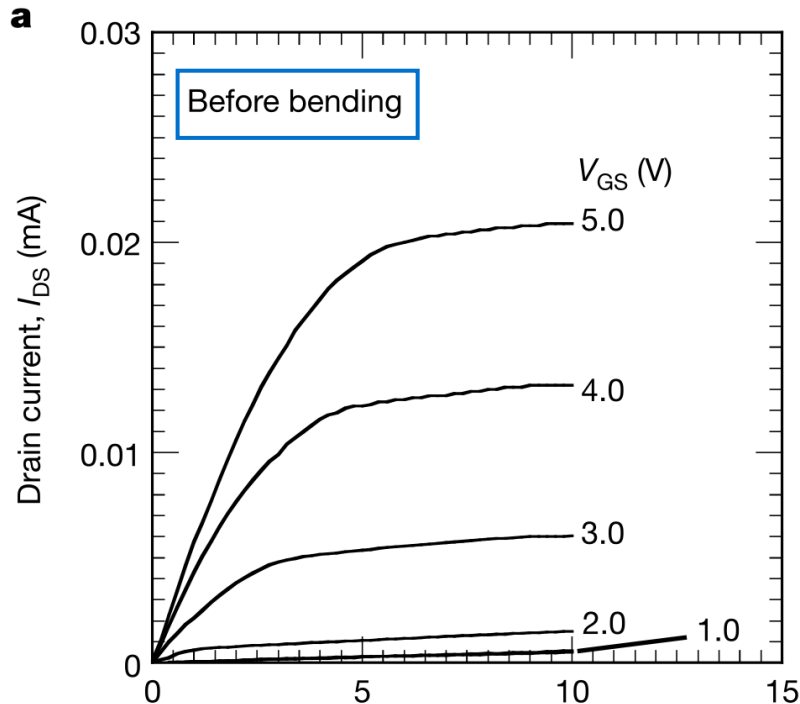
IGZO: Manufacturing

- Pulsed laser deposition (PLD) or Metal-organic chemical vapour deposition (MOCVD) at room (or low) temperature

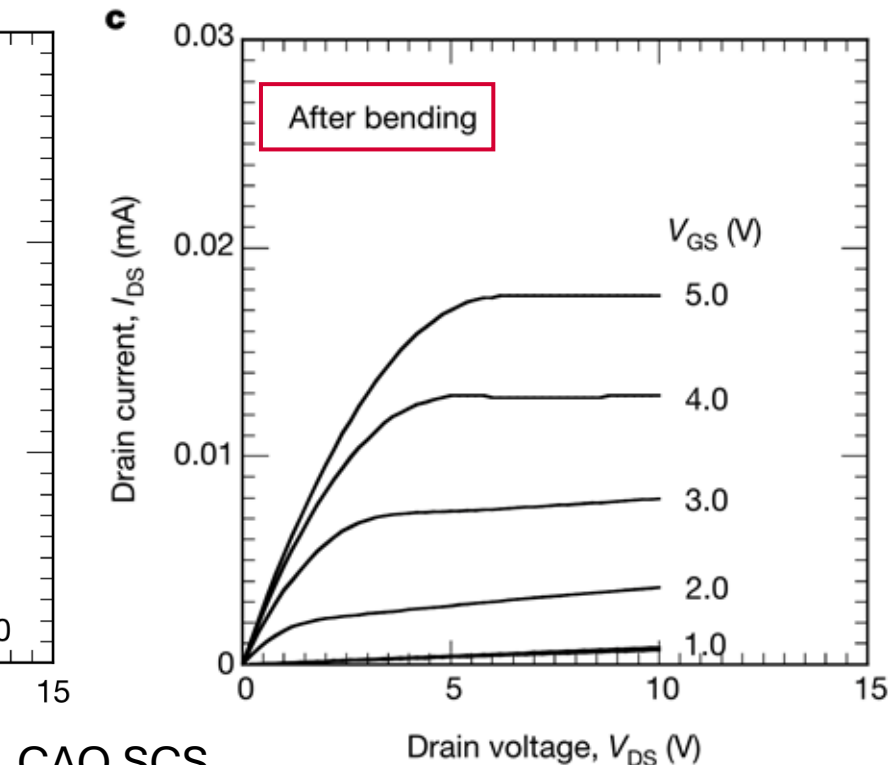


Film thicknesses
a-IGZO active layer : 30 nm
 Y_2O_3 gate : 140 nm
ITO electrode : 40 nm

Key:
 Y_2O_3 : Yttrium oxide
ITO: indium Tin Oxide



Threshold voltage ~ 1.6 V



CAO SCS

Challenge:
How this structure would affect design (upstream)?

Breakout Group Discussions

- Take turns to discuss your answers to either questions:
"Have you heard of any semiconductor materials other than Silicon?
What are the pros and cons of the material?"
"What would be your dream semiconductor material? What
applications will find its best use?"
- Time: 15 minutes

Material	Pros	Cons	Applications

Questions and Answers

- Do you have any questions from today's discussion or the Canvas materials?